

AMENDMENT NO. 2 JANUARY 2023

TO

IS 17261 : 2022 TEXTILES — POLYESTER CONTINUOUS FILAMENT FULLY DRAWN YARNS — SPECIFICATION (*First Revision*)

(Page 1, Clause 3.2) — Substitute the following for existing:

‘3.2 Commercial Allowance — A defined percentage to be added to the net mass of the yarns at ambient conditions for the calculation of commercial mass and certain other properties.

NOTE — The commercial allowance for polyester FDY shall be 3.0 percent.’

(Page 1, Clause 3.3) — Substitute the following for existing:

‘3.3 Commercial Mass — The mass obtained by adding to the net mass at ambient conditions, the mass corresponding to the commercial allowance.’

(Page 1, Clause 3.9) — Substitute the following for existing:

‘3.9 Mass

3.9.1 Gross Mass

The total mass of a consignment, or of any part of a consignment, including the yarn and the tare.

3.9.2 Net Mass

The difference between the gross mass and the corresponding tare, determined at the same time.

3.9.3 Tare

The sum of the masses of all the packing materials such as pallets, wrappers and covers, tie bands/straps, paper tubes, etc, relating to a consignment or to any part of a consignment.’

(Page 3, Clause 4.4.3) — Substitute ‘Bright (BRT)/Optically White (OW)’ for ‘Bright (BRT)/Optically Bright (OBRT)’.

(Page 3, Clause 4.5.3) — Substitute ‘Dope Dyed (DD)’ for ‘Dope Dyed (DD)/Optically White (OW)’.

[Page 5, Table 2, Sl No. (xviii), col (2) and col (3)] — Substitute the following for existing:

‘Recycled yarn RFDY’.

(Page 5, Table 2) — Insert the followings row after Sl No. xix):

‘xx) Virgin yarn VFDY’.

[Page 5, Table 2, Sl No. (vii), col (3)] — Substitute ‘Optically White (OW)’ for ‘Optically Bright (OBRT)’.

[Page 6, Table 3, col (10) and (11), Method of Test] — Substitute the following for existing:

‘Annex G or IS 17087’.

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(Page 8, Clause 6.5) — Insert the following after sentence 1:

‘The requirements of colour fastness to rubbing and colour fastness to perspiration (acidic and alkaline) shall be optional.’

(Page 8, Clause 6.6) — Substitute the following for existing:

‘6.6 Commercial Mass

The commercial mass shall be obtained by adding mass corresponding to commercial allowance of 3.0 percent to the net mass of the consignment at ambient conditions and it shall not be less than the declared commercial mass of the consignment.’

[Page 8, Table 5, Sl No. (iii), col (2)] — Substitute ‘Bright (BRT)/Optically White (OW)’ for ‘Bright (BRT)/OBRT’.

[Page 8, Clause 8.1 (c)] — Substitute ‘Gross or net mass of each carton/pallet;’ for ‘Commercial mass of each carton/pallet’.

[Page 9, Table 6, Sl No. (iii), col (2)] — Substitute ‘Rubbing (optional)’ for ‘Rubbing’.

[Page 9, Table 6, Sl No. (iv), col (2)] — Substitute ‘perspiration (acidic and alkaline) (optional)’ for ‘perspiration (acidic and alkaline)’.

(Page 11, Annex A) — Delete the following reference:

‘IS 7703 (Part 3) : 1991 Methods of Test for continuous filament polyester and polyamide flat yarn — Commercial mass (first revision)’.

(Page 12, Annex B-3) — Insert the following note under the clause:

‘NOTE — Conditioning of samples for at least 4 h is sufficient to attain moisture equilibrium.’

(Page 16, Annex F) — Insert the following Annexure after Annex F and renumber the subsequent annexure:

‘ANNEX G (Table 3)

PROCEDURE FOR DETERMINATION OF SHRINKAGE IN BOILING WATER

G-1 INTRODUCTION

This method is used to determine the residual shrinkage of drawn yarns in skein form when immersed in boiling water for specified time. All measurements shall be made when the yarn is at standard conditions.

G-2 PRINCIPLE

Shrinkage of yarn in skein form is determined when skein is subjected to boiling water for specified time. Shrinkage is expressed as a percentage of original skein length prior to treatment. The loop length of a specimen of conditioned skein of yarn is measured under a specified tension, which is sufficient to straighten but not stretch the skein. The tension free skein is then immersed in boiling water and its loop length is measured again. The shrinkage is calculated as the change in length expressed as a percentage of the length before immersion or exposure of specimen.

G-3 APPARATUS

G-3.1 Hot Water Bath with capacity to completely immerse the magazine.

G-3.2 Magazine to hold the hanks/skeins.

G-3.3 Weights of 50 g to 500 g.

G-3.4 Stainless Steel Scale of one metre length.

G-3.5 Wrap reel to prepare skein.

G-3.6 Metal hooks to take lengths L_1 and L_2 readings.

G-4 PROCEDURE

G-4.1 Skein Preparation

Strip approximately 10 g to 20 g from the surface of the sample.

G-4.2 Prepare the sample skein by calculating the required skein revolutions (R) as follows:

$$R = (2\ 500)/(\text{Denier of sample})$$

NOTE — The number of wraps may be increased or decreased for very fine or very coarse yarns by agreement of the interested parties.

G-4.3 Reel the yarn as a skein using a uniform tension of not over 0.1 gf/den. Lay the yarn smoothly on the reel, overlap and loosely tie the beginning and trailing ends of the skein. To reduce tangling, the skein may be laced loosely. Overlap ends and tie in knot.

G-4.4 Condition the prepared skeins in the standard atmosphere of $(27 \pm 2)^\circ\text{C}$ temperature and relative humidity of (65 ± 2) percent for a minimum of 120 min which is sufficient time for yarn recovery from package winding tensions.

G-4.5 Hang the skein on the top hook and gently add appropriate tensioning weight including hook.

G-4.6 Calculate a tension force in gf corresponding to 0.05 gf/den as follows:

$$\text{Tension force, gf} = 0.05 \times N \times D,$$

where

N = number of ends; that is twice the number of wraps in the skein; and

D = yarn number in denier.

G-4.7 Measure the length of the skein to the nearest mm and record as L_1 . Take measurement within 30 s of placing weight on skein. Do not twist the skein. Make all skein loop length measurements in the standard atmosphere for testing

G-4.8 Remove the tension weight.

G-4.9 Place the skein in the magazine. Repeat the exercise for remaining skeins.

G-4.10 Heat the water bath having demineralized water up to boiling point of water. Immerse the skein magazine in hot demineralized water. Cover/dip all the samples with water. Tie each skein carefully to prevent the entanglement of the yarn in boiling water.

G-4.11 Bring the water bath to a continuously rolling boil and immerse the skeins for 30 min. Immerse the magazine in hot water bath for (30 ± 1) min.

G-4.12 After 30 min, remove the magazine and condition the magazine at room temperature for 60 min. After conditioning, record the length of skein as L_2 with weights same as weights for L_1 .

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G-4.13 Calculation the shrinkage, percent as follows:

$$\text{Shrinkage, Percent} = \frac{(L_1 - L_2)}{L_1} \times 100$$

G-4.14 2 samples shall be taken from each package and test 10 samples. The average of all the readings shall be reported.'

(Page 17, Annex G) — Substitute 'ANNEX H' for 'ANNEX G'.